BCA-202: Discrete Mathematics

Weightage: 70 Marks

Q1. Define a Vector Space.

A set of vectors that can be added together and multiplied by scalars, satisfying eight specific axioms.

$$v + w \in V, kv \in V$$

Q2. What are Groups in Algebraic Structures?

A set with an operation that satisfies closure, associativity, identity, and inverse properties.

$$(G, *)$$

Q3. Explain Rings and Fields.

A Ring has two operations (+, *). A Field is a ring where every non-zero element has a multiplicative inverse.

$$F = (S, +, *)$$

Q4. Define Linear Independence of Vectors.

A set of vectors is linearly independent if no vector in the set can be written as a linear combination of others.

$$c1v1 + c2v2 + ... = 0 \Rightarrow ci = 0$$

Q5. What is a Linear Transformation?

A mapping between two vector spaces that preserves vector addition and scalar multiplication.

$$T(u+v) = T(u) + T(v)$$

Q6. Explain Eigenvalues and Eigenvectors.

For a matrix A, an eigenvector v changes only by a scalar factor λ when multiplied by A.

$$Av = \lambda v$$

Q7. Define Bilinear Forms.

A mapping from two vector spaces to a scalar that is linear in each argument.

$$B(u, v)$$

Q8. What is a Graph in Discrete Math?

A structure consisting of a set of vertices (nodes) and a set of edges (links) connecting them.

$$G = (V, E)$$

Q9. Define Vertex Degree.

The number of edges incident to a vertex in a graph.

$$deg(v)$$

Q10. Explain Paths and Circuits.

A path is a sequence of edges. A circuit is a path that starts and ends at the same vertex.

$$v0 \rightarrow v1 \rightarrow \ldots \rightarrow v0$$

Q11. What is a Planar Graph?

A graph that can be drawn in a plane without any edges crossing each other.

$$V - E + F = 2$$

Q12. Define a Tree in graph theory.

A connected graph that contains no cycles.

$$Edges = Vertices - 1$$

Q13. What is a Rooted Tree?

A tree in which one specific vertex is designated as the root.

Parent-Child Hierarchy

Q14. Explain Spanning Trees.

A subgraph that is a tree and includes all the vertices of the original graph.

$$T \subseteq G$$

Q15. What are Cut Sets?

A set of edges whose removal increases the number of connected components in a graph.

$$G - S$$

Q16. Define Graph Colouring.

Assigning colors to vertices such that no two adjacent vertices have the same color.

$$\chi(G)$$

Q17. What is a Directed Graph (Digraph)?

A graph where edges have a specific direction associated with them.

$$u \rightarrow v$$

Q18. Explain Weighted Graphs.

A graph where each edge is assigned a numerical value or weight.

$$w(e)$$

Q19. What are Multigraphs?

Graphs that are allowed to have multiple edges between the same pair of vertices.

Parallel Edges

Q20. Define Bipartite Graphs.

A graph whose vertices can be divided into two disjoint sets such that no two vertices within the same set are adjacent.

$$V = V1 \cup V2$$